

CAP776:PROGRAMMING IN PYTHON

L:3 T:0 P:2 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: define fundamental concepts of Python including syntax, data types, operators, and basic programming constructs.

CO2 :: understand the working of control structures, functions, data structures, and object-oriented concepts in Python.

CO3 :: apply Python programming concepts, including file handling and libraries such as NumPy and Pandas to solve computational and data-related problems.

CO4 :: analyze data and images using visualization tools like Matplotlib and Seaborn, and basic image processing techniques to interpret results.

Unit I

Basics of Python : introduction python IDLE, jupyter notebook and google colab, creating saving and executing a python file, user Input/Output operations, numeric data types, operators, conditional statement, python loops, user define functions

Unit II

Sequence data types and associated operations : strings, lists and tuples

Other data types and associated operations : dictionary, sets, range

OOP concepts : OOP features, encapsulation, inheritance

Unit III

File handling and Exception handling : built-in exception, try and except, user defined exception, text files: file read/write operation, read JSON file

Unit IV

Data Analysis with NumPy and Pandas : NumPy ndarray object, dimensions in arrays, access array elements, NumPy array manipulation, N-dimensional array(ndarray), understand pandas fundamentals, pandas with csv and html data, pandas operations, pandas with functions, data cleaning process

Unit V

Data visualization : introduction to matplotlib, line plot, bar chart, scatter plot, pie charts, box and whisker plot, histogram, multiple subplots in one figure, introduction to seaborn, seaborn Vs matplotlib, data visualization using seaborn, introduction to data visualization tools to create dashboard

Unit VI

Image Processing using Python : Introduction to digital image processing, Basic Image Operations i.e. Crop, Resize, Rotate, Flip, Adjust brightness, Contrast, Color, Filtering and Enhancement, Blur, Sharpening, Edge detection, OpenCV basics, PIL (Pillow)

List of Practicals / Experiments:

Practical

- Write programs to demonstrate basic concepts of Python including variables, data types, input/output operations, and arithmetic expressions.
- Implement programs using operators and control structures (if, if-else, nested if) to solve decision-making problems.
- Implement programs using looping constructs (for, while) to perform iterative computations, series generation, and pattern printing.
- Implement programs using user-defined functions and recursion for solving computational problems.
- Implement programs using data structures (lists, tuples, sets, and dictionaries) to perform various operations and applications.
- Perform file handling operations including reading, writing, and appending data, along with implementation of exception handling techniques.
- Write programs using NumPy for array creation, indexing, slicing, and numerical computations.

- Perform data manipulation and analysis using Pandas DataFrames.
- Implement programs for data visualization using Matplotlib and Seaborn to represent data graphically.
- Implement image processing operations such as cropping, resizing, rotation, flipping, brightness/contrast adjustment, blurring, sharpening, and edge detection using Python libraries.

Text Books:

1. PROGRAMMING AND PROBLEM SOLVING WITH PYTHON by ASHOK KAMTHANE, AMIT ASHOK KAMTHANE, M.G.Hills

References:

1. PYTHON: THE COMPLETE REFERENCE by MARTIN C. BROWN, MC GRAW HILL
2. DATA SCIENCE AND MACHINE LEARNING USING PYTHON by REEMA THAREJA, MC GRAW HILL